

CROSS-LAYER SECURITY

Research Concept

Cross-Layer Security Metadata Binding and Protocol Analysis
for Adaptive Cyberattack Detection and Prevention

NETWORK	APPLICATION	FILESYSTEM	MEMORY
↓	↓	↓	↓
VALIDATE	CORRELATE	REVALIDATE	DECIDE

Core concept: Give a data object a trusted security identity and follow its security status from the moment it enters the network until it reaches an application, filesystem or memory.

1. WHY IS THIS NEEDED?

A normal security system may inspect a packet, a file and a running process using different security tools. The problem is that these tools may not know that all of them belong to the **same data object**.

TODAY	PROPOSED APPROACH
Network check	Network check + Security ID
File check	File check + same Security ID
Process check	Process check + same Security ID
Memory check	Memory check + same Security ID

The research goal is to connect these observations and create one continuous security story for the object.

2. COMPLETE SECURITY FLOW

STEP	WHAT HAPPENS
1. INCOMING DATA	Packet or data enters the protected system.
2. SECURITY ID	A trusted component assigns a Security Object ID (SOID).
3. HEADER	Important security metadata is associated with the object.
4. CROSS-LAYER CHECK	Network, application, file and memory information are correlated.
5. RISK DECISION	The system determines whether the object is safe, suspicious or malicious.
6. ENFORCEMENT	Allow, monitor, revalidate, quarantine or drop.
7. CONTINUOUS CHECK	Security status is checked again when important conditions change.

3. PROPOSED SECURITY HEADER

Cross-Layer Security Validation Header (CLSVH)

The header is not intended to store the complete file or packet. It carries compact information that tells security components **which object is being checked and what security state applies to it.**

FIELD	WHY IT IS USED
Security Object ID (SOID)	Connects the same logical object across layers.
Parent Object ID	Shows the source or object from which it was derived.
Protocol / Layer	Identifies where the information was observed.
Security State	Shows Unknown, Validated, Suspicious, Quarantined or Malicious.
Risk Score	Provides a value for adaptive security decisions.
Validation Lifetime	Shows whether previous validation is still fresh.
Integrity / MAC	Protects important security metadata from unauthorized modification.

Detailed evidence can remain in a trusted security metadata table while the header carries only the information needed for correlation and fast decisions.

4. HOW THE SECURITY ID CONNECTS EVERYTHING

The **Security Object ID (SOID)** acts like a common reference number for the same logical object.

NETWORK	APPLICATION	FILESYSTEM	MEMORY
SOID-123	SOID-123	SOID-123	SOID-123
Packet evidence	Process evidence	File evidence	Memory evidence

Example: a document arrives through the network. The system can later connect that network event with the process that opened the document and the file stored on disk.

This makes it possible to detect a situation where one layer says “**valid**” but another layer shows suspicious or inconsistent behavior.

5. CROSS-LAYER CRYPTOGRAPHIC LINK

Each layer can add information to a cryptographic chain. This makes it harder for an attacker to secretly replace or alter the security history of the object.

NETWORK	→	APPLICATION	→	FILE	→	MEMORY
C1	→	C2	→	C3	→	C4

Concept: each new validation is linked to the previous validation and the new layer's evidence.

A simple checksum is not enough for attacker-resistant protection. The prototype should evaluate HMAC or digital signatures for authenticated binding.

6. SECURITY VALIDATION LIFETIME

A previous security decision should not automatically remain trusted forever. **SVL** represents how long that validation remains useful.

SECURITY EVENT	RESULT
Validation time expires	Revalidate
File is modified	Previous validation becomes invalid
Process changes	Revalidate
Privilege changes	Revalidate
Protocol / security zone changes	Revalidate
Suspicious behavior appears	Increase inspection

Example: **VALIDATED** → **FILE MODIFIED** → **INVALID** → **RECHECK** → **VALIDATED / QUARANTINED**

7. SECURITY-GATED FILE ADMISSION

The key idea is that network-originated data should not automatically become an accepted persistent file.

CHECK	QUESTION
1. Structure	Is the object a valid file/data structure?
2. Integrity	Has the object changed unexpectedly?
3. Security policy	Is this object allowed?
4. Provenance	Where did it come from and what happened to it?

PASS	FAIL
Filesystem admission	Quarantine / DROP
Continue processing	Stop deeper processing

A trusted security component could issue a **Security Admission Token (SAT)** only after successful validation. The sender must not be allowed to declare itself trusted.

8. ADAPTIVE FIREWALL DECISION

The firewall does not need to perform expensive analysis on every object. It can use a fast first check and deeper inspection only when necessary.

RISK	ACTION	REASON
LOW	ALLOW	No strong indication of danger.
MEDIUM	REVALIDATE / INSPECT	More evidence is required.
HIGH	QUARANTINE	Prevent further processing while investigating.
MALICIOUS	DROP	Stop the object before deeper processing.

This adaptive approach is important because deep inspection, sandboxing and behavior analysis can require significantly more processing resources.

9. FIREWALL CAPACITY

For the research prototype, capacity should be measured in several ways instead of using only one number.

MEASUREMENT	EXAMPLE RESEARCH TARGET
Throughput	1–10 Gbps prototype
Packet processing	1–5 million packets/sec
Connections	100,000–1 million scale
Fast decision latency	Target below 1 ms
Security objects/sec	Measure header validation + correlation rate
Quarantine storage	Based on traffic volume and retention period

Important: 1 Gbps of continuous traffic is about 10.8 TB/day. Therefore, storing all traffic is usually impractical; store metadata and selected suspicious objects instead.

10. HOW TO BUILD THE FIRST PROTOTYPE

PHASE	WORK
1	Capture PCAP traffic and extract packet headers.
2	Analyze L2/L3/L4/L7 protocol relationships.
3	Generate a Security Object ID.
4	Create the proposed security metadata/header.
5	Build integrity and cross-layer correlation checks.
6	Create a simple risk-scoring and policy engine.
7	Test ALLOW / REVALIDATE / QUARANTINE / DROP.
8	Connect network events with processes and files on Linux.
9	Measure detection accuracy, false positives and performance.

Tools: Wireshark / tshark • Zeek • Python • Linux
Future: eBPF/XDP • P4 • SmartNIC / dedicated hardware

11. WHAT SHOULD BE TESTED?

TEST AREA	QUESTION
Detection	Can cross-layer inconsistencies reveal attacks?
Integrity	Can unauthorized changes to security metadata be detected?
Provenance	Can the object be traced from network to host?
SVL	Does event-based revalidation improve security freshness?
Performance	What is the cost in CPU, RAM and latency?
Scalability	How much traffic and how many objects can be processed?
Accuracy	What are detection rate, false positives and false negatives?

12. EXPECTED RESEARCH CONTRIBUTION

The research is not simply proposing another firewall or another IDS. The proposed contribution is a **unified cross-layer security process**.

NETWORK	APPLICATION	FILESYSTEM	MEMORY
Identity	Identity	Identity	Identity
Protocol	Process	File	Execution
Risk	Risk	Risk	Risk

The system combines **identity + provenance + integrity + validation lifetime + cross-layer correlation + adaptive enforcement**.

The novelty of this exact combination must be established through a systematic literature, standards and patent review.

13. SIMPLE EXAMPLE

Scenario: A user receives a document from the network.

STAGE	SYSTEM SEES
Network	Document arrives with network/protocol information.
Security ID	Trusted component assigns SOID.
File	File structure and integrity are checked.
Application	A process opens the file; behavior is monitored.
Memory	Execution-related state is checked if required.
Decision	If evidence becomes suspicious → revalidate / quarantine / drop.

The important point is that all these observations are connected to the **same logical object**.

14. ONE-MINUTE PRESENTATION

“My research proposes a cross-layer security mechanism that follows data from the network to the application, filesystem and memory. The data receives a trusted Security Object ID and security metadata. The system correlates protocol information, integrity, provenance and behavior instead of checking each layer independently. If the object is safe, it continues. If it becomes suspicious, the system revalidates or performs deeper inspection. If it is malicious, it can be quarantined or dropped. A Security Validation Lifetime ensures that old validation does not remain trusted after important changes. The main goal is to detect attacks that may look normal when only one layer is examined.”

WORKING TITLE

Cross-Layer Security Metadata Binding and Protocol Analysis for
Adaptive Cyberattack Detection and Prevention

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IDENTIFY → CORRELATE → VALIDATE → DECIDE